Reflection

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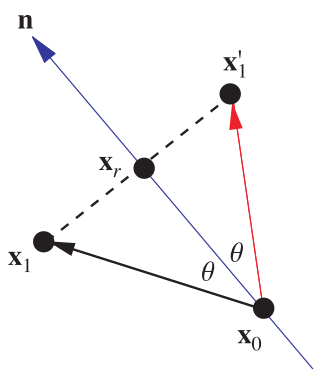
The operation of exchanging all points of a mathematical object with their mirror images (i.e., reflections in a mirror). Objects that do not change handedness under reflection are said to be amphichiral; those that do are said to be chiral.

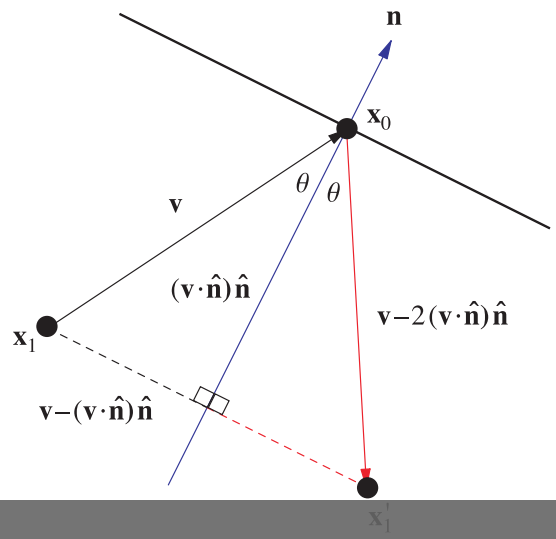


Consider the geometry of the left figure in which a point $\mathbf{x}_1$ is reflected *in* a mirror (blue line). Then

$$\mathbf{x}_r = \mathbf{x}_0 + \hat{\mathbf{n}}\,[(\mathbf{x}_1 - \mathbf{x}_0)\cdot\hat{\mathbf{n}}], \tag{1}$$

so the reflection of $\mathbf{x}_1$ is given by

$$\mathbf{x}_1{}' = -\mathbf{x}_1 + 2\,\mathbf{x}_0 + 2\,\hat{\mathbf{n}}\,[(\mathbf{x}_1 - \mathbf{x}_0)\cdot\hat{\mathbf{n}}]. \tag{2}$$



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normal vector $\mathbf{n}$ satisfies

$$\mathbf{x}_1{}' - \mathbf{x}_0 = \mathbf{v} - 2\,(\mathbf{v}\cdot\hat{\mathbf{n}})\,\hat{\mathbf{n}}. \tag{3}$$

If the plane of reflection is taken as the $y\,z$-plane, the reflection in two- or three-dimensional space consists of making the transformation $x \to -x$ for each point. Consider an arbitrary point $\mathbf{x}_0$ and a plane specified by the equation

$$a\,x + b\,y + c\,z + d = 0. \tag{4}$$

This plane has normal vector

$$\mathbf{n} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}, \tag{5}$$

and the signed point-plane distance is

$$D = \frac{a\,x_0 + b\,y_0 + c\,z_0 + d}{\sqrt{a^2 + b^2 + c^2}}. \tag{6}$$

The position of the point reflected in the given plane is therefore given by

$$\mathbf{x}_0{}' = \mathbf{x}_0 - 2\,D\,\hat{\mathbf{n}} \tag{7}$$

$$= \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix} - \frac{2\,(a\,x_0 + b\,y_0 + c\,z_0 + d)}{a^2 + b^2 + c^2} \begin{bmatrix} a \\ b \\ c \end{bmatrix}. \tag{8}$$

The reflection of a point with trilinear coordinates $\alpha_0 : \beta_0 : \gamma_0$ in a point $\alpha_1 : \beta_1 : \gamma_1$ is given by $\alpha : \beta : \gamma$, where

$$\alpha = 2\,\alpha_1\,(b\,\beta_0 + c\,\gamma_0) + \alpha_0\,(a\,\alpha_1 - b\,\beta_1 - c\,\gamma_1) \tag{9}$$

$$\beta = 2\,\beta_1\,(a\,\alpha_0 + c\,\gamma_0) + \beta_0\,(-a\,\alpha_1 + b\,\beta_1 - c\,\gamma_1) \tag{10}$$

$$\gamma = 2\,\gamma_1\,(a\,\alpha_0 + b\,\beta_0) + \gamma_0\,(-a\,\alpha_1 - b\,\beta_1 + c\,\gamma_1). \tag{11}$$

SEE ALSO

Affine Transformation, Amphichiral, Chiral, Dilation, Enantiomer, Expansion, Glide, Handedness, Improper Rotation, Inversion Operation, Mirror Image, Projection, Reflection Triangle, Reflection Property, Reflection Relation, Reflexible, Rotation, Translation

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= expand (x + y + z)^10

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